

# Clicks or Comments?

## The quality-quantity trade-off of review systems

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Matthew J. H. Murphy

University of Pennsylvania

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## Designing review systems: Quality vs. Frequency

- Platforms use reviews to make decisions. How should they design their review systems?

The trade-offs are natural in this context:

- The more detailed the review system, the more information is being extracted from each individual reviewer.
- It is less costly for reviewers to leave a binary review than a full review.
  - This leads to more simple reviews being reported than detailed reviews.

**Is it possible to quantify the trade-off between the informativeness of reviews and how frequently they are submitted?**

Model

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## Model: Reviewers' information

There are two possible values of product quality  $\theta \in \Theta = \{L, H\}$  with prior  $q = \mathbb{P}(L)$ .

- There are  $N$  “reviewers”. Reviewer  $i$  has a (conditionally independent) private signal  $S_i \in \mathbb{R}$  about  $\theta$  (realization  $s_i$ ).

$$s_i = u_\theta + \sigma \epsilon_i, \quad \text{where} \quad u_\theta = \begin{cases} -\mu & \text{if } \theta = L \\ \mu & \text{if } \theta = H \end{cases}$$

- Interpret  $s_i$  as an average utility  $u_\theta$  with some taste shock  $\epsilon_i$  for reviewer  $i$ .
- The noise  $\sigma$  controls the precision of reviews.

### Assumption

The  $\epsilon_i$  are i.i.d., with continuous, full-support, log-concave density given by  $f$  (with CDF  $F$ ).

## Model: The platform's problem & review systems

- The platform has a state-dependent utility function  $u : \mathcal{A} \times \Theta \rightarrow \mathbb{R}$ , with action set  $\mathcal{A}$  finite.
- It chooses a review system  $r : \mathbb{R} \rightarrow \mathbb{R}$  that maps signals to reviews.
  - E.g., the full review system:  $r_f(s) := s$ ; or
  - A binary threshold system:  $r_\tau(s) := \mathbb{1}\{s > \tau\}$ , for some  $\tau \in \mathbb{R}$ .
- Let  $\mathcal{R}_r = \text{supp}(r(S_i))$  denote the set of possible reviews under  $r$ .
- Each review system  $r$  has an associated *probability of review*  $p_r$ , that is decreasing in  $|\mathcal{R}_r|$ .
  - More complicated reviews are less likely to be reported.
  - For today, independent of reviewers' signals.

## Model: Timing

- The platform chooses a review system.
- Reviewers observe private signals  $s_i$  and decide whether to submit.
- The platform observes submitted reviews  $\mathcal{S}$ , updates beliefs to posterior  $\pi(r, \mathcal{S})$  that  $\theta = L$ , and takes an action to maximize its expected utility:

$$u^*(r, \mathcal{S}) := \max_{a \in \mathcal{A}} \{ \pi(r, \mathcal{S})u(a, L) + (1 - \pi(r, \mathcal{S}))u(a, H) \}.$$

- The platform aims to maximize  $u^*(r, N) := \mathbb{E}[u^*(r, \mathcal{S})]$  by choosing  $r$ :

$$\max_r u^*(r, N).$$

- I focus on the case when  $N$  is large, to emulate online reviews.

## Trading off Quality and Frequency

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Each review system  $r$  induces a distribution over reviews in state  $L$  and state  $H$ . Define the measure  $\gamma_L^r$  over  $\mathcal{R}_r$  as, for any measurable  $B \subseteq \mathcal{R}_r$ ,

$$\gamma_L^r(B) := \mathbb{P}_L(r^{-1}(B)) = \int_{s:r(s) \in B} \frac{1}{\sigma} f\left(\frac{s + \mu}{\sigma}\right) ds. \quad \text{Similarly define } \gamma_H^r.$$



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### Definition

Given a review system  $r$ , define its *learning efficiency* as

$$\nu(r) := 1 - \min_{\lambda \in [0,1]} \int_{\mathcal{R}_r} \left( \frac{d\gamma_L^r}{d\gamma_H^r} \right)^\lambda d\gamma_H^r.$$

## Trading off between frequency and quality

- The probability of review  $p_r$  also impacts platform's learning.

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### Theorem

Fix a decision problem and two review systems  $(r, p_r)$  and  $(r', p_{r'})$ . If

$$p_r \nu(r) > p_{r'} \nu(r')$$

there exists an  $\bar{N}$  such that for all  $N > \bar{N}$ ,  $u^*(r, N) > u^*(r', N)$ .

- Implies that platform separably trades off between quality and frequency of review systems.

## The relative information of review systems

- Can re-frame the result as

$$p_r \nu(r) > p_{r'} \nu(r') \iff \frac{\nu(r)}{\nu(r')} > \frac{p_{r'}}{p_r}.$$

- Since platforms can compute  $p_r$  for different review systems, this suggests a focus on the ratio  $\nu(r)/\nu(r')$ .
- Specifically, can focus on the information contained in a review system relative to the full review system  $r_f(s) = s$ .

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## Definition

Given a review system  $r$ , define its *relative information* as

$$\kappa(r) := \frac{\nu(r)}{\nu(r_f)} \in [0, 1].$$

## Imprecise Signals and the Optimal Binary Review System

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## Imprecise signals

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- Normalize signals  $s \mapsto \frac{s}{\sigma}$  to simplify notation.

Goal: characterize the optimal binary review system when signals are imprecise, and determine when it outperforms the full review system.



# Characterizing the optimal binary review system

Call  $r$  a *threshold system* if  $r(s) := \mathbb{1}\{s > \tau\}$  for some  $\tau \in \mathbb{R}$ . Denote by  $r_\tau$ .

## Theorem

The optimal binary review system when signals are imprecise is given by the threshold review system  $r_{\tau^*}(s) := \mathbb{1}\{s > \tau^*\}$

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$$\tau^* = \operatorname{argmax}_{\tau} \left\{ \frac{f(\tau)^2}{F(\tau)} + \frac{f(\tau)^2}{1 - F(\tau)} \right\}.$$

This review system is preferred to the full review system if

$$\underbrace{\left( \frac{f(\tau^*)^2}{F(\tau^*)} + \frac{f(\tau^*)^2}{1 - F(\tau^*)} \right) \cdot \frac{1}{I_f}}_{\text{relative information}} > \underbrace{\frac{p_{r_f}}{p_{r_{\tau^*}}}}_{\text{ratio of review frequencies}},$$

where  $I_f := \mathbb{E} \left[ \left( \frac{f'(s)}{f(s)} \right)^2 \right]$  denotes the Fisher Information of  $f$ .

## Taste Heterogeneity

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I next study the properties of the optimal binary review system as a function of reviewer taste heterogeneity.

- Heterogeneous tastes  $\Rightarrow$  extreme tastes are common (e.g., movies).
- Homogeneous tastes  $\Rightarrow$  extreme tastes are uncommon (e.g., toasters).

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When reviewers are heterogeneous, their idiosyncratic tastes have larger tails.

- I focus on a specific functional form for  $f$ :

$$f_{\alpha}(x) \propto \exp(-|x|^{\alpha}), \quad \text{for } \alpha \geq 1.$$

## Illustration of different review systems

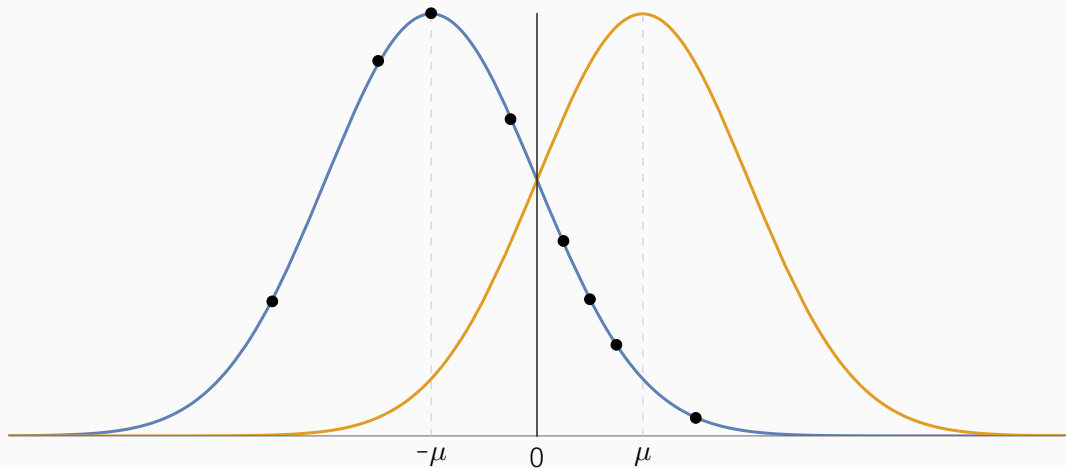


Figure 1: Illustration of possible released signals.

# Illustration of different review systems

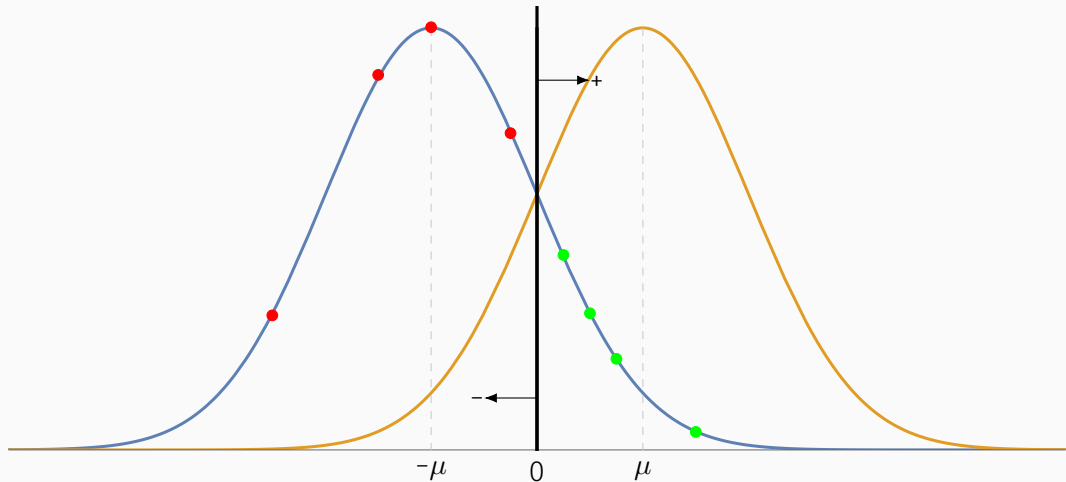


Figure 1: Illustration of the “good”/“bad” review system  $r_0$ .



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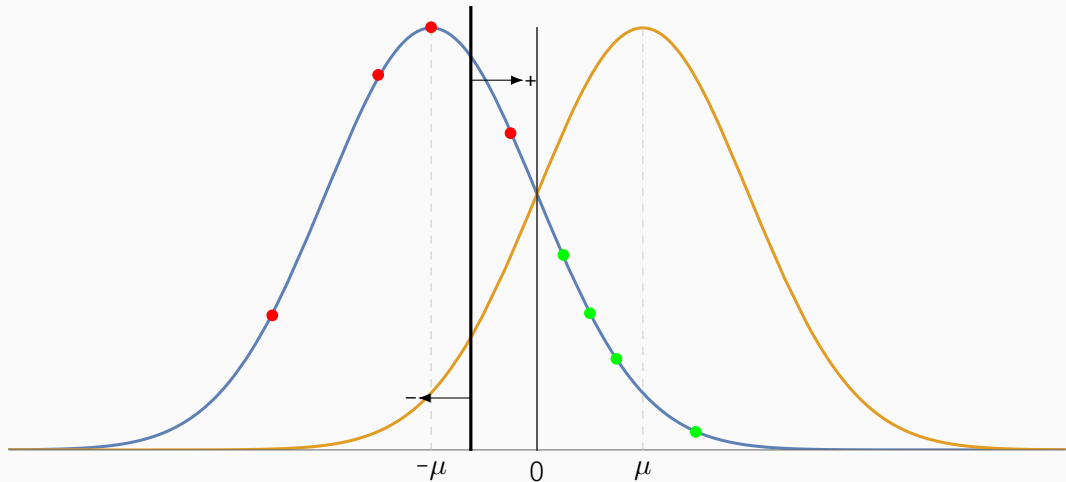


Figure 1: Illustration of the “horrible” review system  $r_\tau$ ,  $\tau < 0$ .

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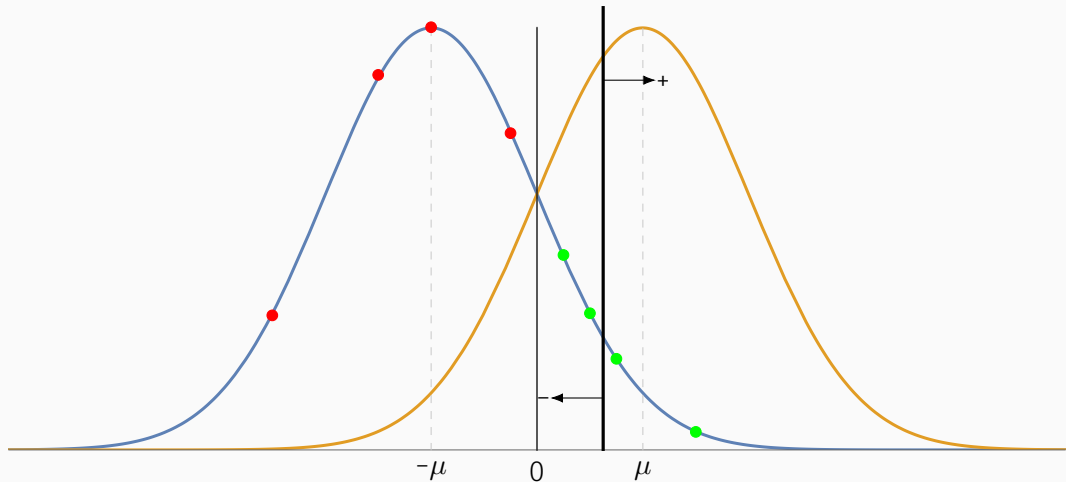
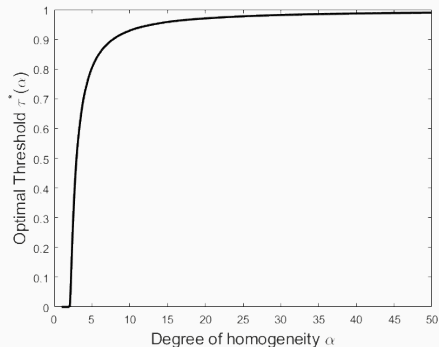


Figure 1: Illustration of the “amazing” review system  $r_\tau$ ,  $\tau > 0$ .

# The optimal binary review system

## Proposition

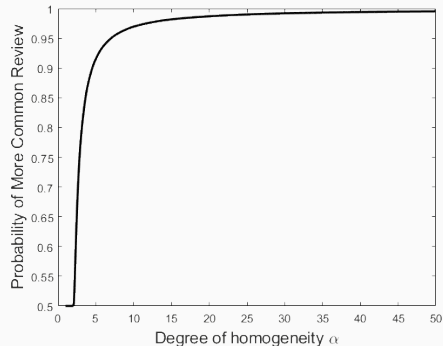
- (i) The “good”/“bad” review system  $r_0$  is optimal iff tastes are very heterogeneous ( $\alpha \leq 2$ ).



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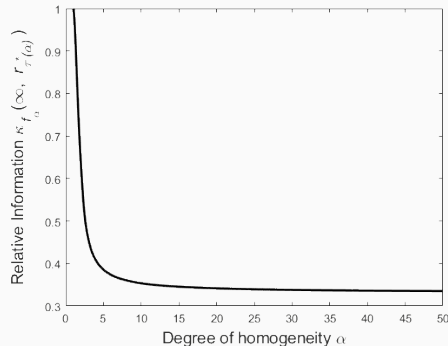
- (i) The “good”/“bad” review system  $r_0$  is optimal iff tastes are very heterogeneous ( $\alpha \leq 2$ ).
- (ii) As tastes become more homogeneous ( $\alpha$  increases), the optimal review system is more “horrible” or “amazing”.



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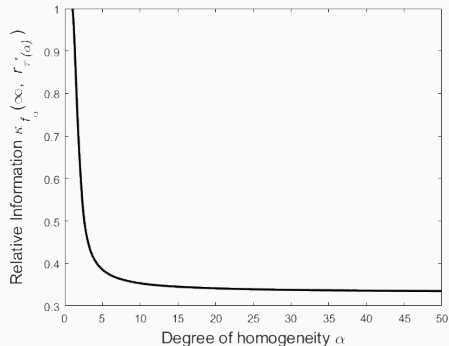
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- (ii) As tastes become more homogeneous ( $\alpha$  increases), the optimal review system is more “horrible” or “amazing”.
- (iii) The optimal binary review system is preferred to the full review system if  $p_{r_\tau} > 3.25p_{r_f}$ , for all  $\alpha$ .



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- (i) The “good”/“bad” review system  $r_0$  is optimal iff tastes are very heterogeneous ( $\alpha \leq 2$ ).
- (ii) As tastes become more homogeneous ( $\alpha$  increases), the optimal review system is more “horrible” or “amazing”.
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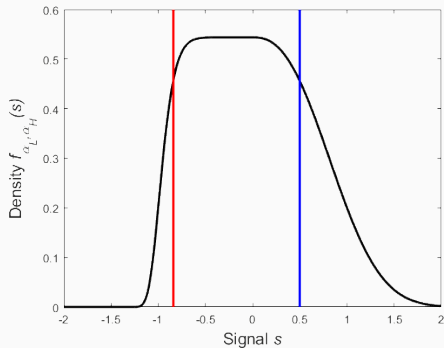
Both the “horrible” and “amazing” asymmetric review systems are optimal.

# When to use the “horrible” review system over the “amazing”

If one side of the distribution is more homogeneous than the other, the platform should always set the threshold to isolate the homogeneous side.

## Proposition

If negative signals are more homogeneous than positive signals ( $\alpha_L > \alpha_H$ ), the optimal binary review system is “horrible”.



# Conclusion

- I study how a platform designs optimal review systems, trading off between the informativeness of reviews and how frequently they are submitted.
- I study how reviewer heterogeneity affects the performance of simple review systems.
- My results suggest an explanation for why positively-skewed review systems are common in certain settings, and not in others.

In the paper:

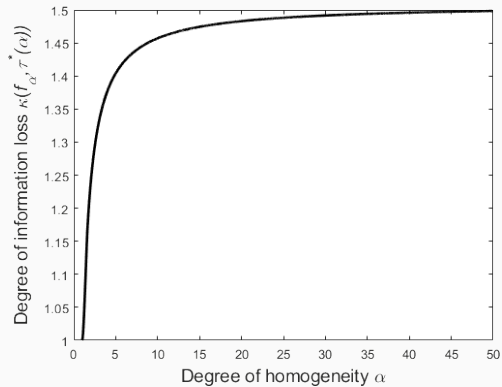
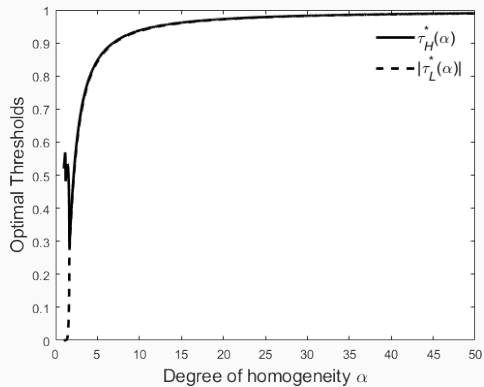
- Beyond binary review systems.
- Dependence of reporting rates on reviewers' signals.



## Extensions

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### 3 Bin Comparative Statics



## Non-Uniform Reporting Rates: Notation

Suppose now that the probability that a reviewer decides to leave a review is given by

$$p(r, s) = p_r q(s).$$

Now, define the distribution of reviews in state  $\theta$  as

$$\gamma_L^{(p,r)}(B) := p_r \int_{s:r(s) \in B} q(s) f_\theta(s) ds$$

Then the generalization of  $p_r \nu(r)$  is given by

$$\rho(\mu; r, p) := 1 - \min_{\lambda \in [0,1]} \left[ \int_{R_r} \left( d\gamma_L^{(p,r)} \right)^\lambda \left( d\gamma_H^{(p,r)} \right)^{1-\lambda} + \left( \gamma_L^{(1-p,r)}(\mathbb{R}) \right)^\lambda \left( \gamma_H^{(1-p,r)}(\mathbb{R}) \right)^{1-\lambda} \right].$$

## Generalization of Learning loss near Uninformative Signals

$$E(g) := \int_{-\infty}^{\infty} g(s)q(s)f(s)ds \quad \text{and} \quad E(g; \tilde{r}) := \int_{-\infty}^{\infty} g(s)\mathbb{1}\{r(s) \in \tilde{r}\}q(s)f(s)ds$$

### Theorem

Suppose that  $p(r, s) = p_r q(s)$  where  $q$  is continuous. Let  $\ell_f := \frac{f'}{f}$  denote the linear score of  $f$ . Then learning-loss for a finite review system  $r$  is given by

$$\frac{p_r}{p_f} \cdot \lim_{\mu \rightarrow 0} \frac{\rho(\mu; p_f \cdot q)}{\rho_q(\mu; r, p_r \cdot q)} = \frac{E(\ell_f^2) + \frac{p_f E(\ell_f)^2}{1 - p_f E(1)}}{\sum_{\tilde{r} \in R_r} \frac{E(\ell_f; \tilde{r})^2}{E(1; \tilde{r})} + \frac{p_r E(\ell_f)^2}{1 - p_r E(1)}}.$$

# Null Reviews are informative, but not as $p_r \rightarrow 0$

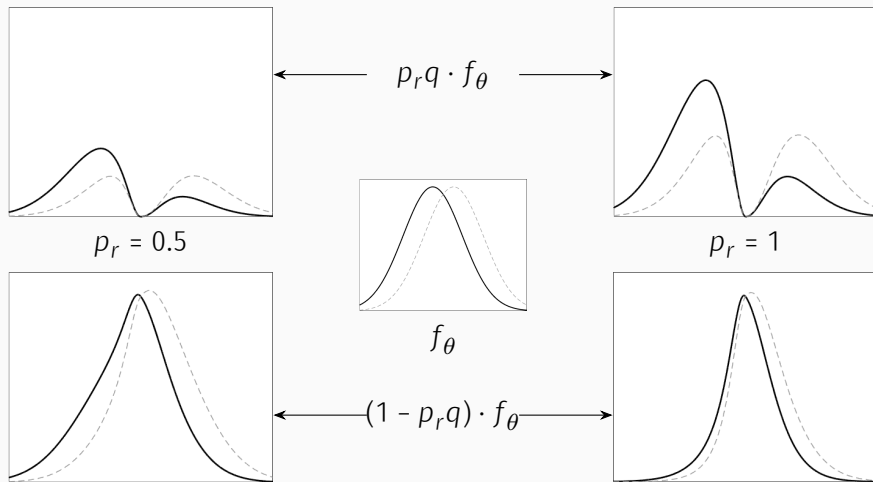


Figure 3: Example with  $q(s) = 1 - \frac{1}{1+2(x^2+4\mathbb{1}\{x \leq 0\}x^2)}$ .

We have thus far focused on the decision of the platform when eliciting information from reviewers. What if this information has already been collected?

### Lemma

Fix a consumer with a decision problem, and two review systems  $r, r'$ . Then there exists an  $n$  large such that for all  $n_r, n_{r'} > n(\mu)$ ,  $n_r$  reviews of type  $r$  is preferred to  $n_{r'}$  reviews of type  $r'$  if

$$\frac{n_r}{n_{r'}} > \frac{\log(1 - \nu(\mu; r'))}{\log(1 - \nu(\mu; r))}.$$

# Relationship between Production and Consumption of Information

We have the following:

## Lemma

Let  $\zeta(\mu) := \frac{\nu(\mu; r')}{\nu(\mu; r)}$  and let  $\iota(\mu) := \frac{\log(1-\nu(\mu; r'))}{\log(1-\nu(\mu; r))}$ . Then, we have that

$$\iota(\mu) > \zeta(\mu) \quad \text{if and only if} \quad \nu(\mu; r') > \nu(\mu; r).$$

Moreover, we have that

$$\lim_{\mu \rightarrow 0} \zeta(\mu) = \lim_{\mu \rightarrow 0} \iota(\mu).$$